

Efficiently Coloring the Intersection of a General Matroid and Combinatorial Matroids

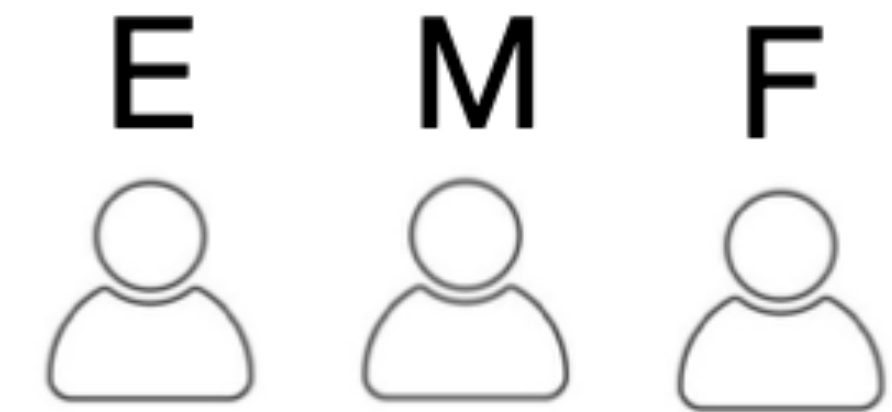
Stephen Arndt, Benjamin Moseley, Kirk Pruhs, and Michael Zlatin

Matroid Basics

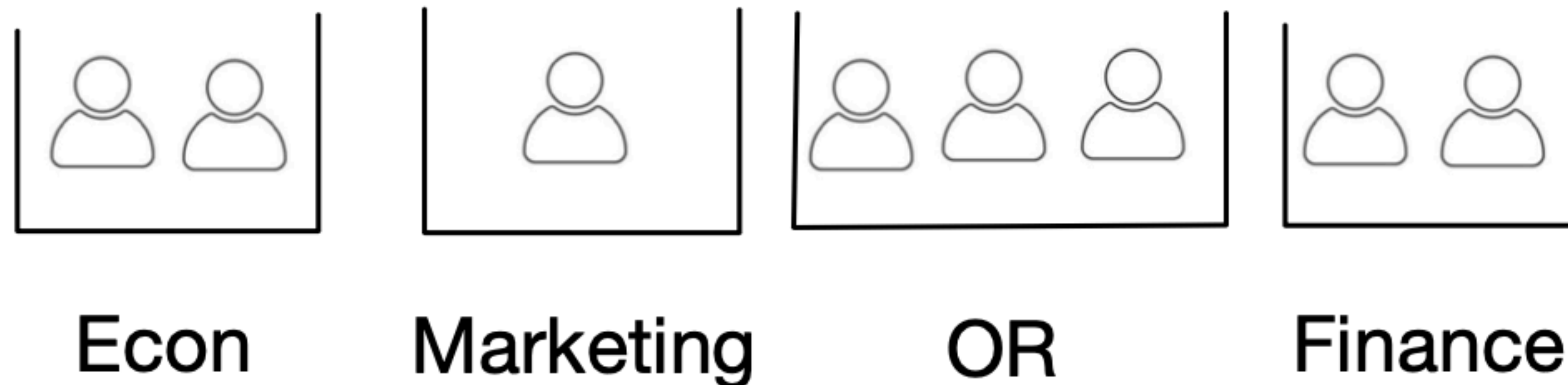
Matroid Basics

- A Motivating Example
- Matroid
- Partition Matroid
- Matroid Intersection
- Chromatic Number

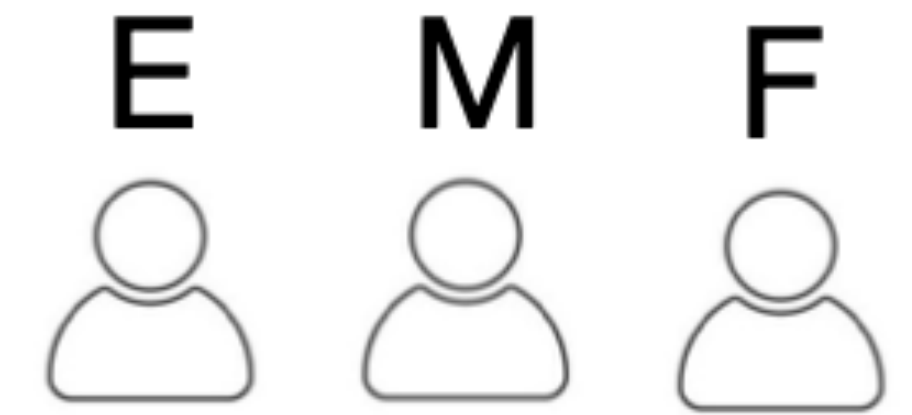
Committees of Faculty



- Faculty in departments
- **Committee:** At most 1 faculty per department
- **Subset Property:** Every subset of a committee is a committee.
- **Exchange Property:** We can always “grow” a smaller committee using a larger committee.



Committees of Faculty

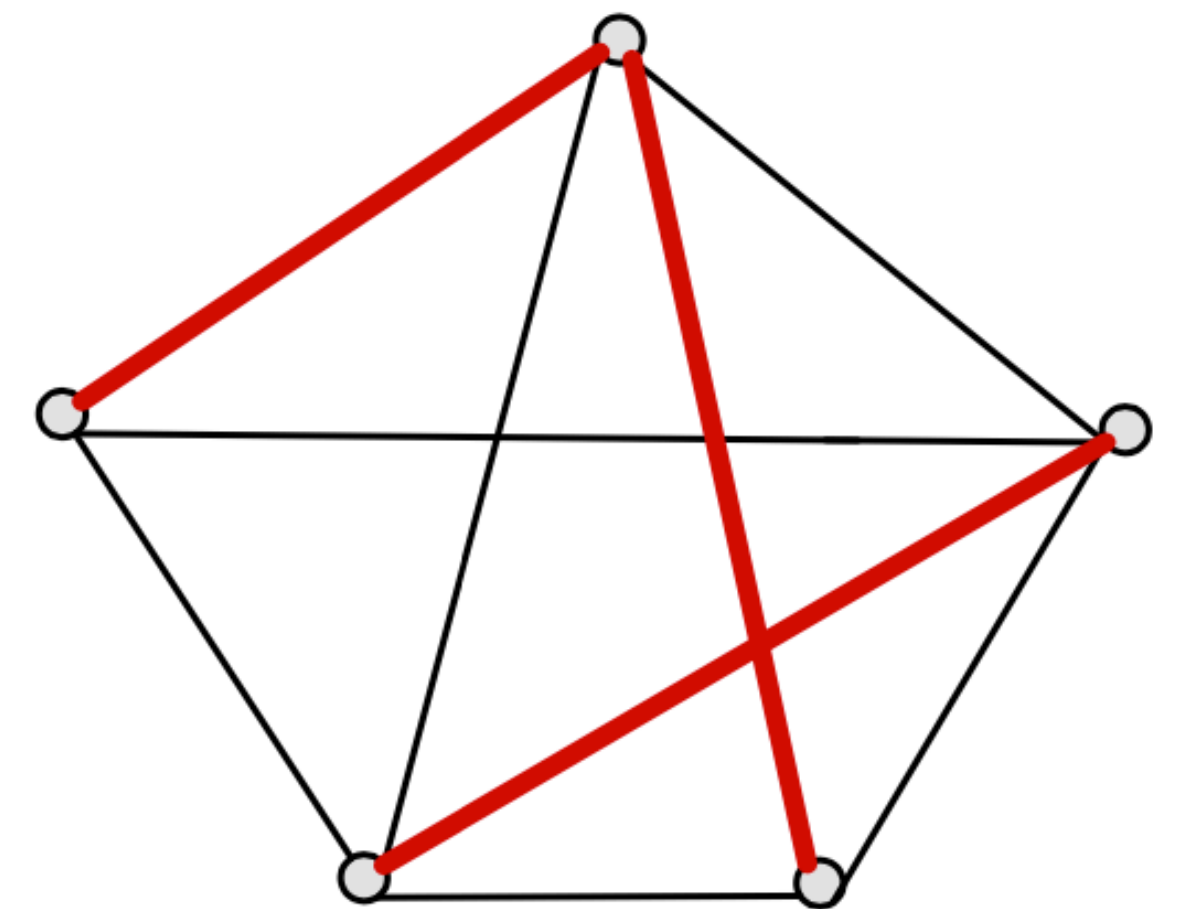
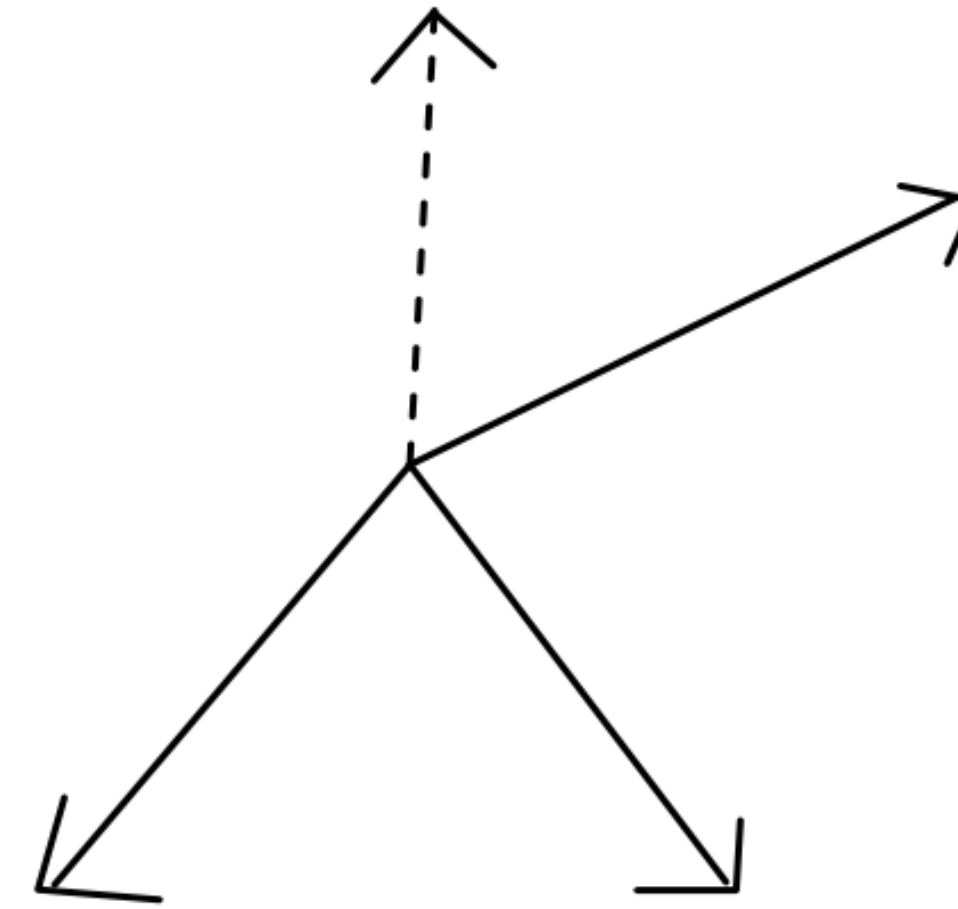


- Faculty in departments
- **Committee:** At most 1 faculty per department
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Matroids

- A **matroid** is a set system $M = (E, \mathcal{S})$ with **subset** and **exchange** properties.
- Linearly independent sets
- Forests in an undirected graph



Matroid Definition

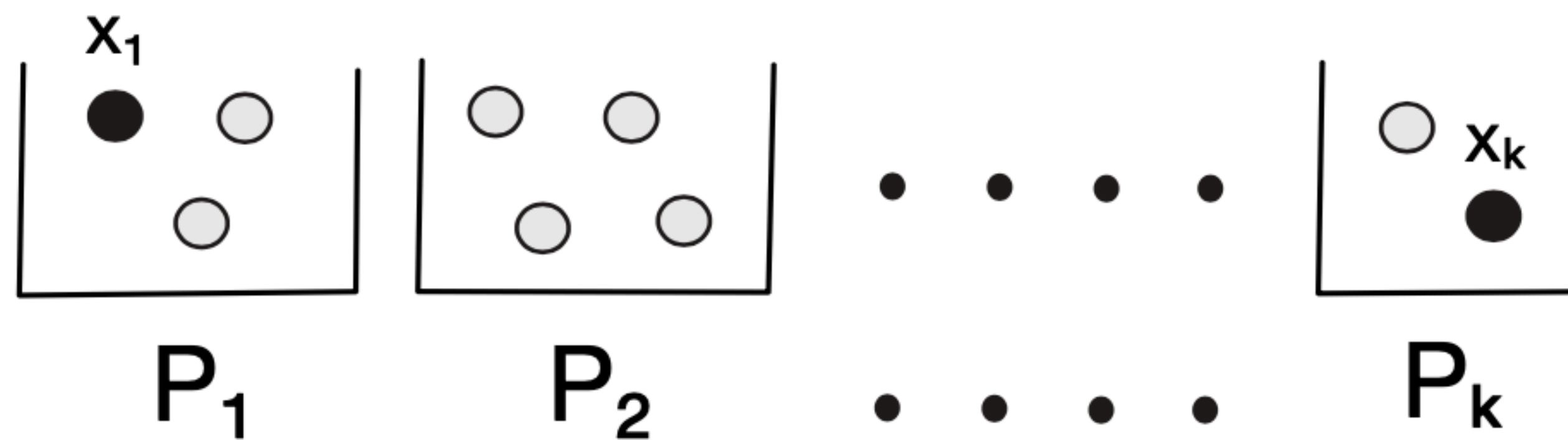
- **Def:** A matroid is $M = (E, \mathcal{S})$ where...
 - E is the set of elements, called the **ground set**.
 - $\mathcal{S}$ is a family of subsets of E , called the **independent sets**.
 - **Subset Property:** If $I \in \mathcal{S}$ and $J \subseteq I$, then $J \in \mathcal{S}$.
 - **Exchange Property:** If $I, J \in \mathcal{S}$ and I is larger than J , then there is some $x \in I \setminus J$ such that $J \cup \{x\} \in \mathcal{S}$.

Importance of Matroids

- **Fundamental objects** in combinatorial optimization
- **Simplify and generalize** combinatorial results
- Extensively studied in the “ACO” literature

Partition Matroid

- Elements E in parts $P_1, P_2, \dots, P_k$
- **Independent Sets:** At most c_i elements per part P_i



$\{x_1, x_4, \dots, x_k\}$ is independent

- Simple and natural type of matroid

Matroid Intersection

- **Given:** Two matroids $M_1 = (E, \mathcal{S}_1)$ and $M_2 = (E, \mathcal{S}_2)$
- **Def:** The matroid intersection $M_1 \cap M_2$ is...

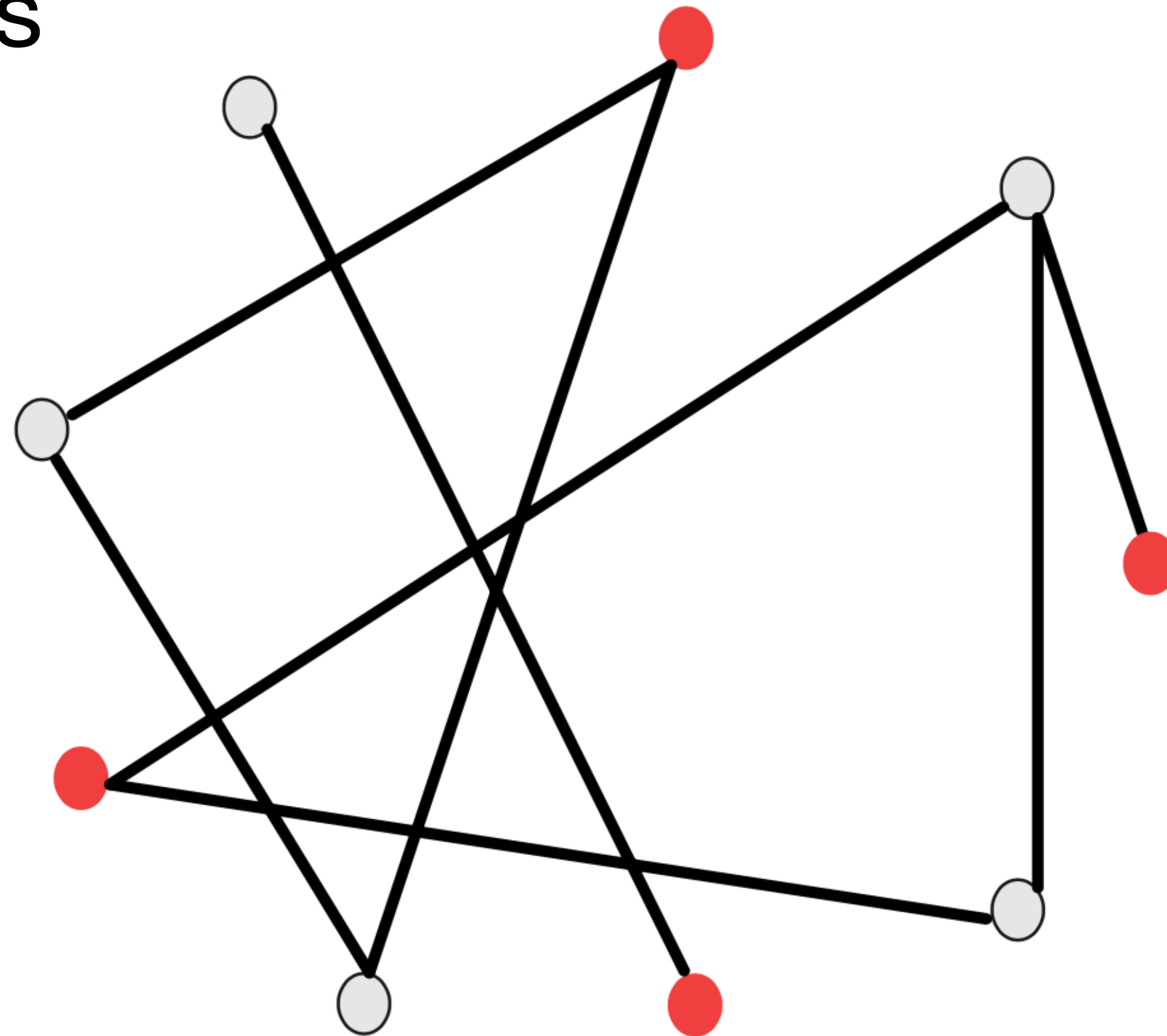
$$M_1 \cap M_2 = (E, \mathcal{S}_1 \cap \mathcal{S}_2)$$

- **Subset property** holds
- **k Matroids:**

$$M_1 \cap M_2 \cap \dots \cap M_k = (E, \mathcal{S}_1 \cap \mathcal{S}_2 \cap \dots \cap \mathcal{S}_k)$$

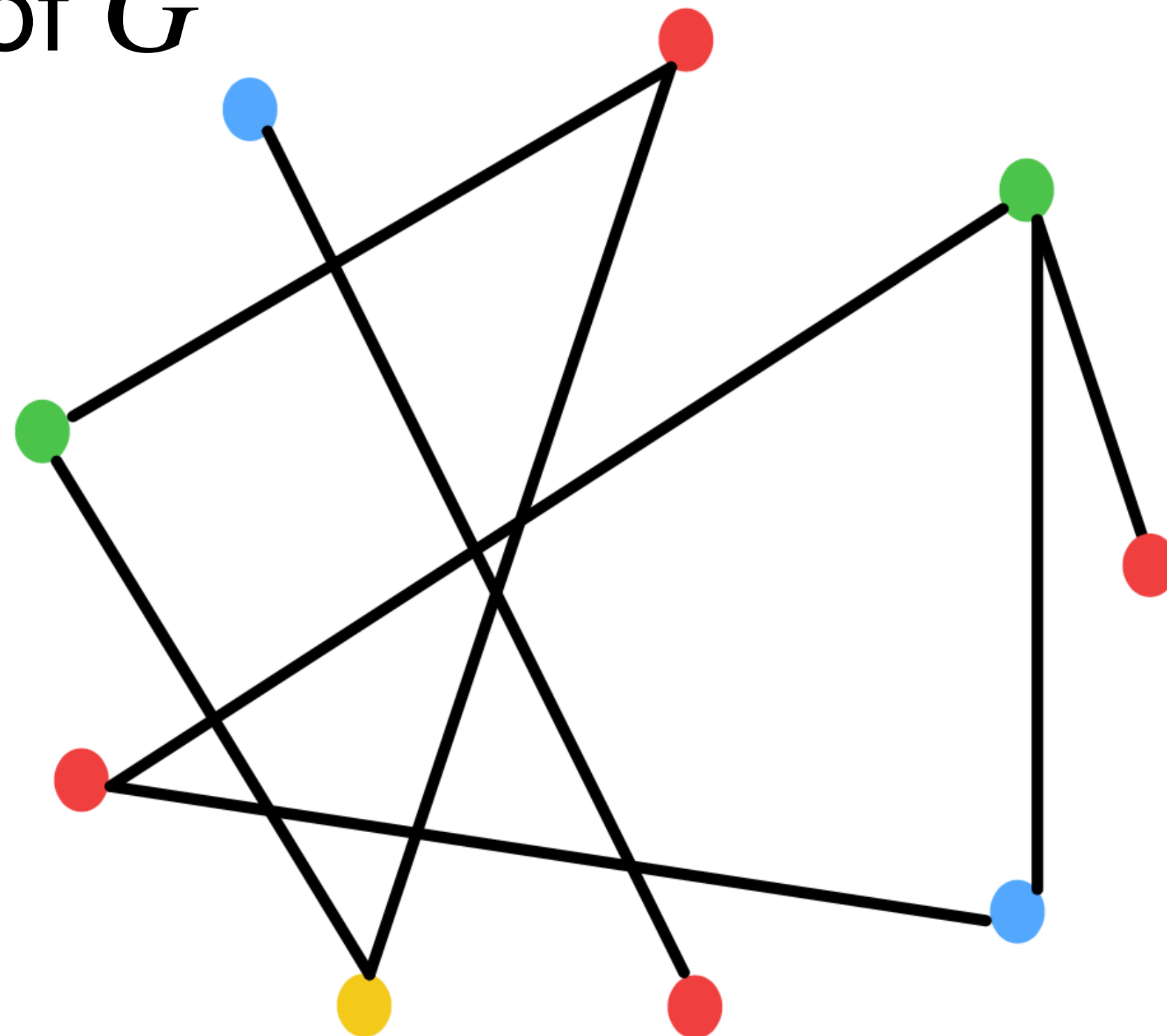
Chromatic Number

- **Given:** A set system $N = (E, \mathcal{S})$ with the **subset property**.
 - **Ex.** Stable sets in a graph G .



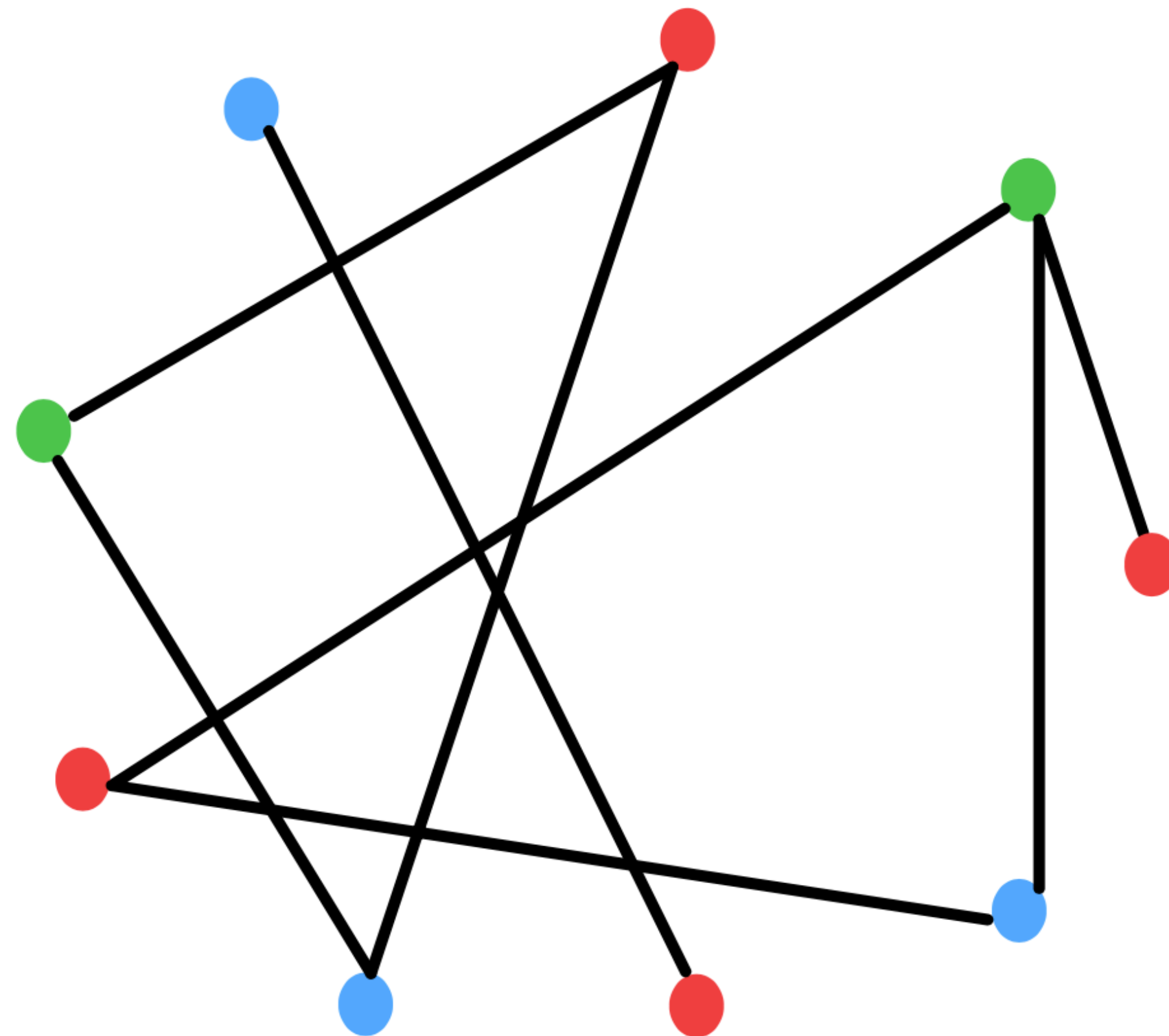
Chromatic Number

- **Def:** A **valid coloring** of $N = (E, \mathcal{S})$ is a coloring of the elements such that each **color class** is in $\mathcal{S}$.
 - **Ex.** 4-coloring of G



Chromatic Number

- **Def:** The **chromatic number** $\chi(N)$ of $N = (E, \mathcal{S})$ is the **minimum** number of colors in any valid coloring.
- **Ex.** $\chi(G) = 3$



Matroid-Intersection Coloring Problem

Matroid-Intersection Coloring Problem

The Problem

- **Given:** Two matroids $M_1 = (E, \mathcal{S}_1)$ and $M_2 = (E, \mathcal{S}_2)$
- **Goal:** Compute a minimum valid coloring of $M_1 \cap M_2$
- **OPT:** $\chi(M_1 \cap M_2)$

Key Results

- **One Matroid:** Polynomial-time solvable [Edm68, Sch03]
- **Two Matroids:** NP-Hard [BS21, BCK23, HKK24]
 - Even if $M_2 =$ Partition Matroid
- Approximation algorithms
 - **Want:** # colors $\leq \alpha \cdot \text{OPT}$.

Our Contribution

Our Contribution

1. **2-approximation** for Matroid-Intersection Coloring Problem when M_1 is **general** matroid and M_2 is **partition** matroid.
 - Deterministic algorithm
 - No $O(1)$ -approximation was known in this setting

Our Contribution

1. **2-approximation** for Matroid-Intersection Coloring Problem when M_1 is **general** matroid and M_2 is **partition** matroid.
2. **k -approximation** for k -MICP when M_1 is **general** matroid and $M_2, M_3, \dots, M_k$ are **partition** matroids.
3. **$(2k - 1)$ -approximation** for k -MICP when M_1 is **general** matroid and $M_2, M_3, \dots, M_k$ are “**combinatorial**” matroids.

Our Contribution

1. **2-approximation** for Matroid-Intersection Coloring Problem when M_1 is **general** matroid and M_2 is **partition** matroid.
- **Main insight:** Extend Edmonds' **augmenting path** algorithm for single-matroid coloring [**Edm68, Sch03**].

Understanding the Literature

Understanding the Literature

- Types of Matroids
 - **Combinatorial:** Partition, Graphic, Transversal, Laminar...
 - **General:** All matroids
- Types of Proofs
 - **Nonconstructive:** Does **not** yield a poly-time algorithm.
 - **Constructive:** Does yield a poly-time algorithm.

Understanding the Literature

Matroids

General

Combinatorial

Nonconstructive

Constructive

Proofs

Understanding the Literature

Matroids

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Understanding the Literature

Matroids

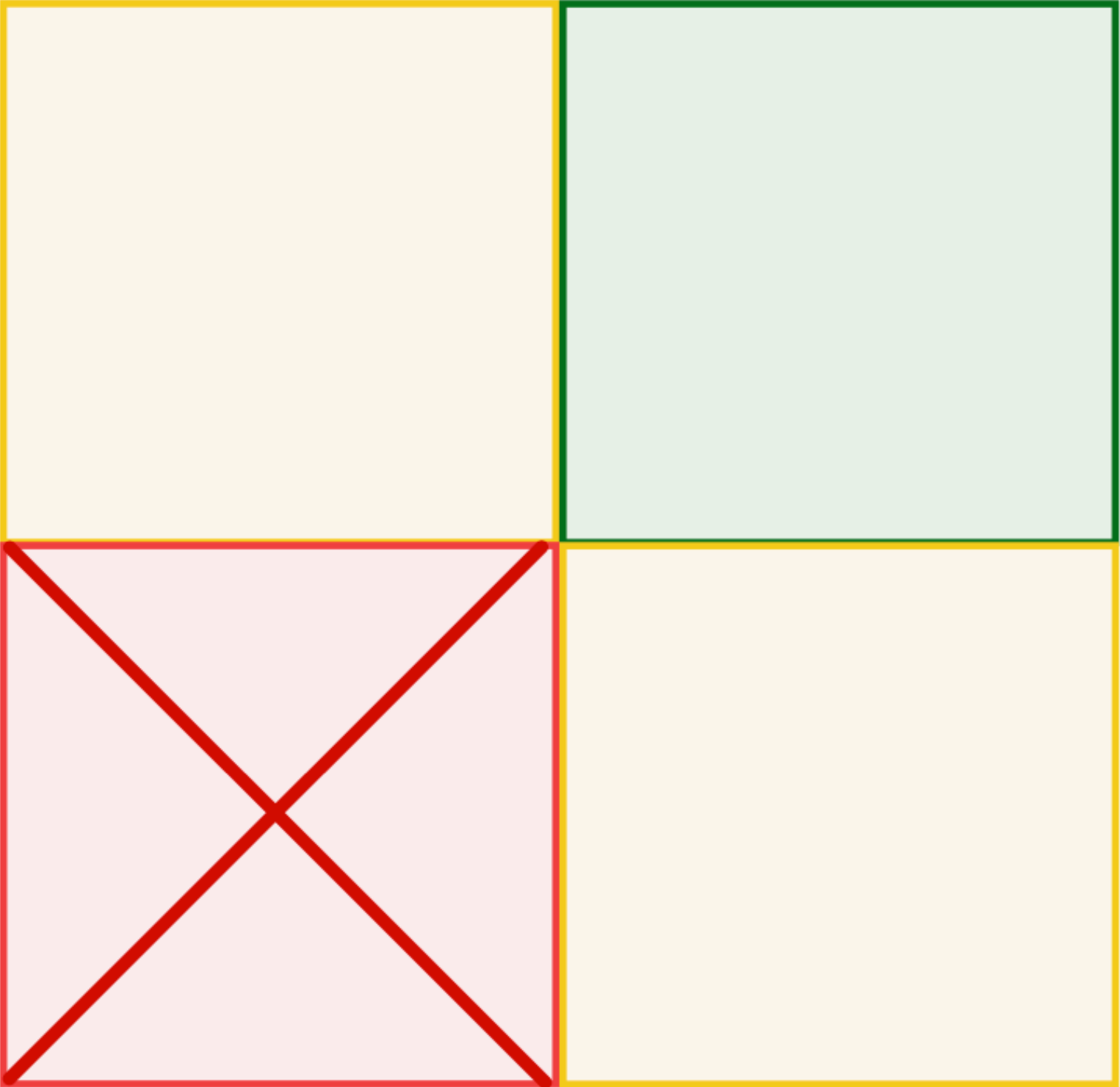
General

Combinatorial

Nonconstructive

Constructive

Proofs

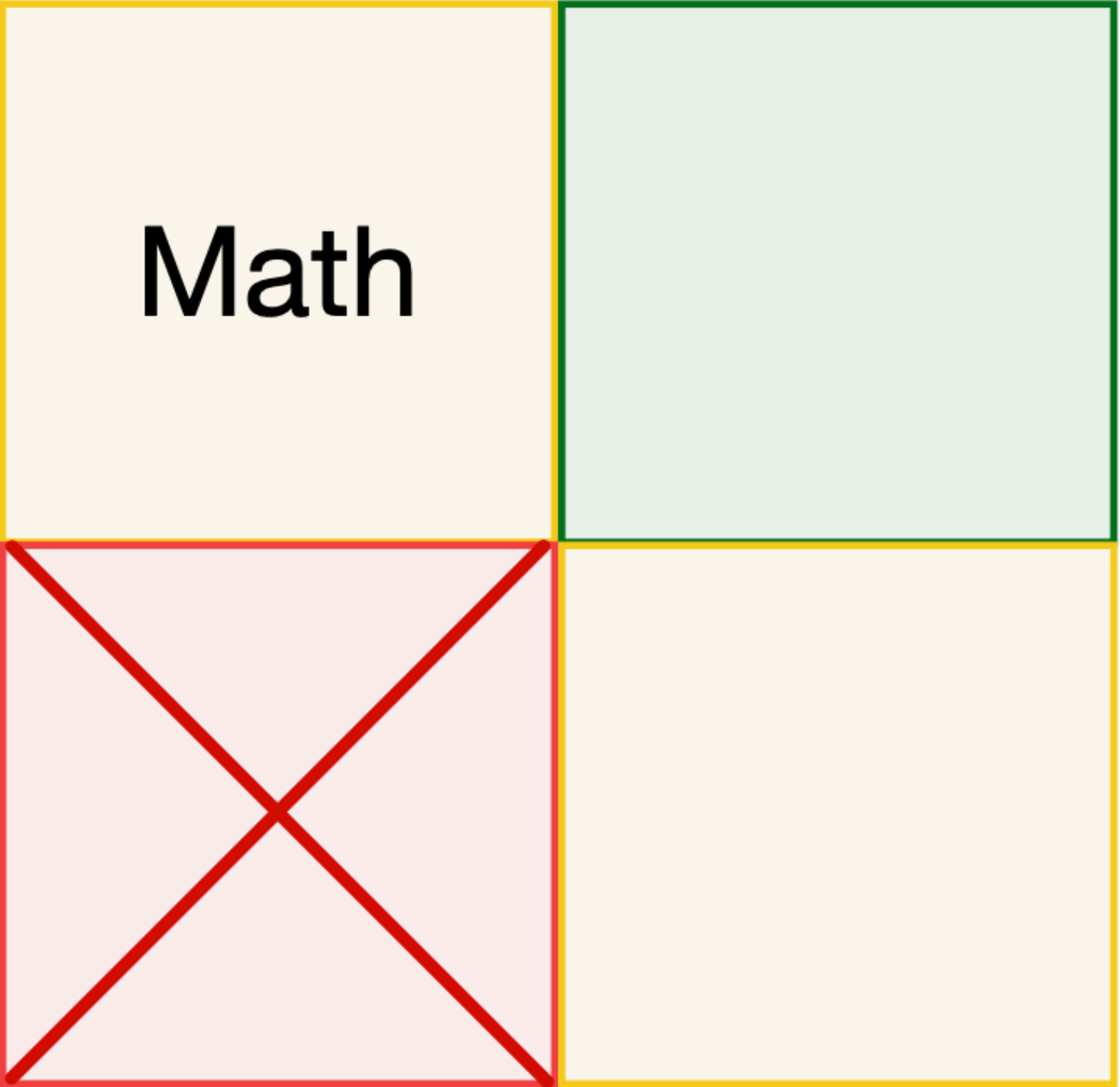


Understanding the Literature

Matroids

General

Combinatorial



Nonconstructive

Constructive

Proofs

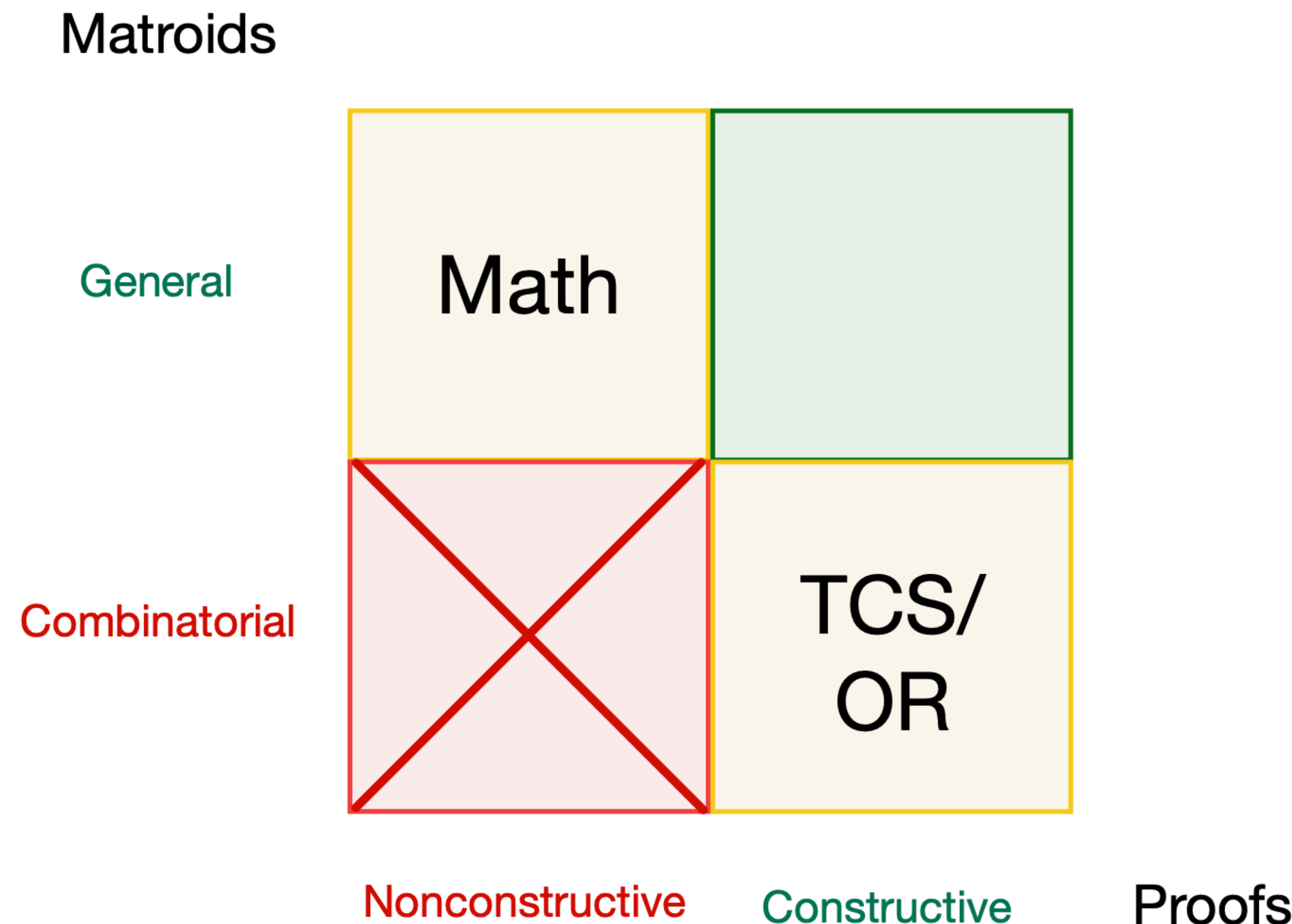
Math: **General** and **Nonconstructive**

- 10-100 papers (many **combinatorial** and **nonconstructive**)
- Thm ([**AB06**, **BG25**]): For two **general** matroids M_1 and M_2 ,

$$\chi(M_1 \cap M_2) \leq \chi(M_1) + \chi(M_2)$$

- Topological methods and Sperner's Lemma
- **Nonconstructive**

Understanding the Literature



TCS/OR: Combinatorial and Constructive

- 5-10 papers
- **Two Partition Matroids ([Kön16])**
 - Polynomial-time algorithm to **exactly** compute $\chi(M_1 \cap M_2)$.
 - Reduce to edge coloring a bipartite graph.

TCS/OR: **Combinatorial** and **Constructive**

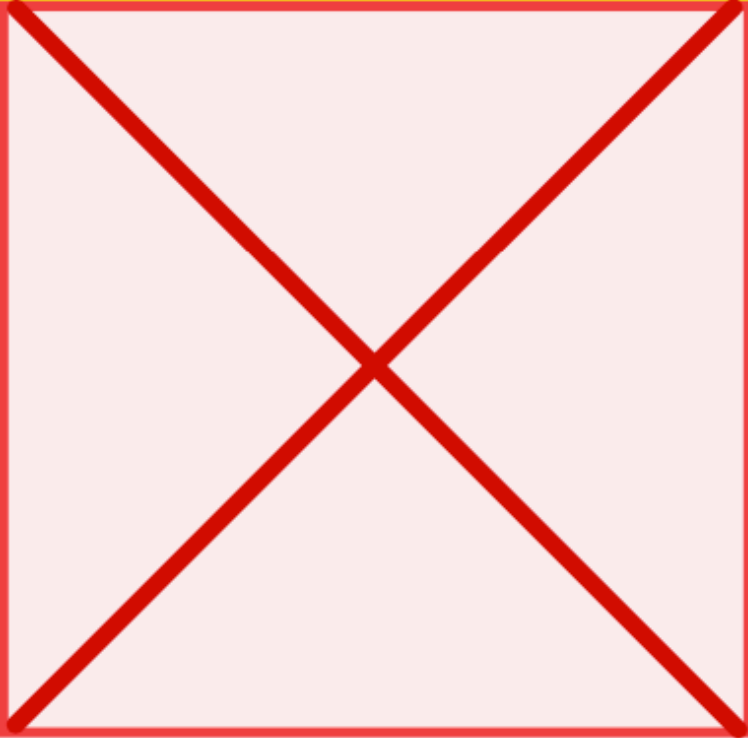
- **Two Combinatorial Matroids** ([IMP21, BSY21, LMP21])
 - Partition reduction technique
 - **Types:** Graphic, Transversal, Laminar, Gammoid...
 - Convert matroid into **partition** matroid with **constant-factor** blowup in the chromatic number.
 - **Conj** ([BSY21]): Every matroid is reducible in this way.
 - **Thm** ([LMP22, AKKG23]): Binary matroids are not.

Understanding the Literature

Matroids

General

Combinatorial

Math	?
	TCS/ OR

Nonconstructive

Constructive

Proofs

?: General and Constructive

- 0 papers
- One **General** + One **Partition** Matroid
 - $O(\log n)$ -approximation algorithm via **Set Cover**
 - Bad approximation factor 😞
 - Does not connect $\chi(M_1 \cap M_2)$ to $\chi(M_1)$ and $\chi(M_2)$ 😞

?: General and Constructive

- **Thm** ([AB06, BG25]): For two **general** matroids M_1 and M_2 ,

$$\chi(M_1 \cap M_2) \leq \chi(M_1) + \chi(M_2)$$

- **Research Question:** Does there exist a **constructive** proof (polynomial-time algorithm) showing

$$\chi(M_1 \cap M_2) \leq \chi(M_1) + \chi(M_2)$$

if M_1 is a **general** matroid and M_2 is a **partition** matroid?

Our Contribution

Our Contribution

- **Thm:** For a **general** matroid M_1 and **partition** matroid M_2 , there is a **polynomial-time** algorithm that computes a

$$(\chi(M_1) + \chi(M_2) - 1)\text{-coloring of } M_1 \cap M_2.$$

- First polynomial-time $O(1)$ -approximation algorithm for MICP using a **general** matroid
- **Constructive** proof of $\chi(M_1 \cap M_2) \leq \chi(M_1) + \chi(M_2) - 1$

Algorithm Overview

Matroid Coloring Problem

Edmonds' Algorithm ([Edm68, Sch03])

- One **General** Matroid
- **Outer Loop:** i elements colored
 - Build exchange graph G
 - Find **color-chordless** augmenting path P
 - Update coloring

Matroid-Intersection Coloring Problem

Our Algorithm

- One **General** Matroid + One **Partition** Matroid
- **Outer Loop:** i elements colored
 - Build exchange graph G of M_1 with $\chi(M_2) - 1$ **extra** colors
 - Find **color-chordless subgraph** H (many **augmenting paths** for M_1)
 - Find **suffix-feasible path** P in H (with respect to M_2)
 - Update coloring

The image is a graphic with a red background. A faint world map is visible in the background. The word "BREAKING" is written in large, white, bold, sans-serif capital letters. It is positioned on a red rectangular background that has a slight 3D effect with a white highlight on the top left corner. To the left of the word "BREAKING", there are three horizontal white lines of increasing length, suggesting motion or a list. Below "BREAKING", the word "NEWS" is written in large, dark blue, bold, sans-serif capital letters. It is positioned on a white rectangular background that also has a slight 3D effect with a blue highlight on the bottom right corner. The overall design is dynamic and urgent, typical of a news broadcast graphic.

BREAKING

NEWS

Recent Developments

New Paper w/ Swamy

1. **2-approximation** for two **general** matroids

Constructive proof of...

$$\chi(M_1 \cap M_2) \leq 2 \max \{ \chi(M_1), \chi(M_2) \}$$

This Paper

1. **2-approximation** for one **general** + one **partition** matroid

Constructive proof of...

$$\chi(M_1 \cap M_2) \leq \chi(M_1) + \chi(M_2) - 1$$

Q1: Constructive proof of $\chi(M_1 \cap M_2) \leq \chi(M_1) + \chi(M_2)$ for two **general** matroids?

?

Recent Developments

New Paper w/ Swamy

2. $(k^2 - k)$ -approximation for k general matroids

Quadratic in k

This Paper

2. k -approximation for one general and $k - 1$ partition matroids

3. $(2k - 1)$ -approximation for one general and $k - 1$ combinatorial matroids

Linear in k

Q2: $O(k)$ -approximation for k general matroids?

Linear in k



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Questions?